

A CNN-Attention Network for Continuous Estimation of Finger Kinematics from Surface Electromyography

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Abstract—Dexterous control of robotic hand driven by human motor intent has drawn a lot of attention in both industrial and rehabilitation scenarios. Providing simultaneous and proportional control has become a prevailing solution recently. Towards improving the finger kinematics estimation precision and reducing its computational cost, a convolution model with attention mechanism (CNN-Attention) was proposed in this study. For comparison purpose, two previously used deep learning models, the long short-term memory (LSTM) and the Sparse Pseudo-input Gaussian processes (SPGP) were also included. By using surface electromyography (sEMG) and kinematic signals corresponding to six hand grasp movements, the estimation performance of each of the three models was evaluated with three measures, Pearson Correlation Coefficient (CC), Root Mean Square Error (RMSE), and coefficient of determination (R²) between real and estimated joint angles. The results demonstrated that the proposed CNN-Attention model outperformed LSTM and SPGP significantly (p -value < 0.05), with an average value of CC, RMSE, and R², 0.87, 9.65 degrees, and 0.73, respectively. Also, the CNN-Attention model is more stable and versatile over various subjects and joint angles in comparison to other models. Additionally, the computational time to build a CNN-Attention was obviously shorter than that to train a LSTM model (43.00 ± 4.25 min vs. 73.40 ± 5.81 min). These findings suggest that the CNN-Attention would be a promising model for continuous estimation of hand movements in the human-machine interaction and cooperation.

Index Terms—sEMG, joint kinematics, continuous estimation, CNN-Attention, deep learning.

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I. INTRODUCTION

OVER decades, robotic hands have been intensively investigated and applied in many scenarios, like search and rescue, space teleoperation, service and industry, as well as artificial prosthetics. Towards providing dexterous hand poses, anthropomorphic robotic hands with kinematic resemblance of mechanical joints and links have been developed [1]. Moreover, great efforts have been made to provide functional resemblance of human-hand functionalities based on the use's motor intention. Surface electromyography (sEMG) which reflects the neuromuscular activities of human beings, is a direct source to decode human motor intention, which has been widely used for dexterous control of robotic hands worldwide [2].

Conventional sEMG-based robotic control methods depend mainly on manually selected features and machine learning algorithms, i.e., the myoelectric pattern recognition (MPR) framework. Towards identifying more complicated motion classes, an increasing number of deep learning approaches have been also proposed and proved to be of potential [2], [3]. However, the main limitation of these classification algorithms is that only discrete recognition of predefined motion classes can be output, and the motor identification performance would degrade with the increase of number of degrees of freedom (DOF). Moreover, only sEMG corresponding to isometric muscle contractions were used to build the MPR model, which limits the fine-grained control even with high offline decoding accuracy, in particular when motion switching with abrupt EMG patterns. To solve the aforementioned shortcomings of MPR framework, continuous motion estimation [4]–[6], i.e., estimating the joint angles or torques of multiple DOF based on sEMG is more promising in the accurate and safe human-machine interaction.

Current continuous motion estimation methods can be generally categorized into two classes, the model-based and the model-free approaches. The model-based approaches depend on the existence of “proven laws”, which need to accurately describe the dynamics of the constitutive units and functional relationship between observed variables [7], and are thus limited by the uncertainties that we cannot measure internal physiological parameters always. Therefore, researchers tend to use model-free approaches when facing practical control problems. For example, a regression-based control approach proposed by Hahne *et al.* allows for simultaneous and proportional control (SPC) of two DOF, which has been evaluated on five prosthetic

end users [8]. Considering the drawbacks of linear regression control scheme that the subjective mapping between sEMG and control outputs would create a redundancy, a series of SPC methods inspired by muscle synergy were proposed for robotic control with two or three DOFs [9]–[11]. However, these muscle-synergy-inspired methods have been rarely used for robotic hand control except in the work by Furui *et al.* where a motion-generation model was proposed [12]. One important reason maybe that the muscle-synergy-inspired approaches used for SPC of two-DOF or three-DOF have a limited generalization ability for more dexterous hand control.

Aiming at providing SPC of robotic hands, an increasing number of literatures began to shift focus on deep learning based continuous motion estimation approaches recently [13], [14], for which require no background knowledge about muscle physiology. For instance, Jimson Ngeo *et al.* proposed a multi-output convolved Gaussian Process to estimate 16 continuous joint angles in hand. They found the Correlation Coefficient (CC) had an obvious difference in different joints, and the average CC was less than 0.8 [15]. To improve the precision of finger kinematics estimation, some new deep learning models and feature extraction methods were proposed recently [14], [16], [17]. Note in practical scenarios with small size of available EMG datasets and sparse data, the deep learning models tend to overfit and end up with poor generalization ability [13]. To solve this problem, Rahimian *et al.* proposed a Few-Shot learning architecture, but which was aimed at gesture classification [3]. Recently, Avian *et al.* employed Manifold Learning to handle sparse data and LSTM based attention mechanism as a regressor, which enhanced the finger joint estimation performance significantly [18], but these methods have a drawback in slower data training. In brief, in addition to the estimation accuracy, the high computational cost to build a deep learning model has been also proved a key challenge and impediment to the clinical deployment for these deep learning approaches.

Therefore, towards improving the finger kinematics estimation precision and reducing the computational cost to train a model in the meanwhile, a convolution with attention mechanism network (CNN-Attention) was proposed in this study. The multi-scale convolution structure with filters of different size was used to represent the sEMG feature by extracting both fine and coarse information [19], and the multi-head attention structure which could learn the long-range dependencies of time sequences more easily was adopted to improve the accuracy and speed up the model training in the meanwhile [20]. It is hypothesized that the CNN-Attention network is more efficient, yielding not only high estimation accuracy but also costing less time to build the model, so as to speed up the practical application of deep learning models in real scenarios.

II. METHODS

A. Database

For easy access, the newly proposed CNN-Attention network was evaluated on Ninapro [21], a publicly available database designed for facilitating the research of artificial intelligence robots and prosthetic hands. Considering the diversity of subjects, the second Ninapro database referred as DB2 was utilized

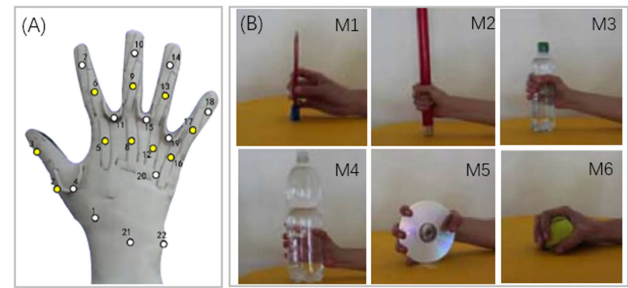


Fig. 1. Dataset Description. (A) The distribution of 10 joint angles denoted as yellow dots, and (B) six types of grasp movements were included in this study.

in this work, which consists of 50 gestures, including wrist, hand, grasping, and functionality exercise and force patterns from 40 healthy subjects. Each movement was repeated 6 times, held for 5 seconds and then rest for 3 seconds. The Trigno Wireless EMG system (Delsys, Inc, Natick, Massachusetts, USA) was used to record the sEMG with 12 wireless electrodes, and the sampling rate was set as 2 kHz. At the same time, the CyberGlove II data-glove was used to recording 22 finger joint angles with a sampling rate of 20 Hz. Herein, we used 12 channels of sEMG to estimate 10 finger joint angles, namely the proximal interphalangeal point and Metacarpophalangeal point as shown in, because they are the main active joints in the grasping movement Fig. 1(A).

In current study, 15 typical subjects were chosen out of the database, for who cover all subjects' information as much as possible, i.e., the range of the height weights, gender, age and dominant hand. For each subject, six types of grasping movements based on representative shapes and diameters were included (Fig. 1(B)). The shapes included a cylinder, a ball and a flat object. The diameters included large, medium and small diameter objects.

B. Data Pre-processing

The sEMG signals were filtered with a Butterworth bandpass filter to remove the direct current (DC) component and high frequency domain noise, and a notch filter was utilized to attenuate the power line interference. Then, a sliding window with length of 100 ms and step of 0.5 ms was used to segment the sEMG into consecutive windows, given the tradeoff between acceptable accuracy and controller delay. Afterwards, the root mean square (RMS) was extracted from each window to construct the sEMG feature matrix [22], [23]. In order to synchronize with sEMG, the finger joint angles were resampled to 2000 Hz. Then a Zero-phase lowpass filter was used to smooth the joint angle signals. In line with the pre-processing of sEMG, the finger joint angle signals were also divided into fragment sequences of 100 ms duration and sliding step-length of 0.5 ms. The last joint angle value within each window was used to construct the joint angle feature matrix.

After separate feature extraction for sEMG and finger joint angles, the obtained feature matrices were normalized to [-1,1]. Given the requirement of data sufficiency for deep learning, the sEMG feature matrix was converted into a multi-channel long exposure EMG sample collection which contains 100 subframes RMS features (one sub-frame RMS feature is calculated from

100 ms EMG and the sliding step length is 0.5 ms). In order to perform train and test experiment on our proposed CNN-Attention network, each movement (containing six trials) was segmented into training dataset and testing dataset, respectively. The training dataset was composed of four trials that were randomly selected from all six trials, and the remaining two trials were designated as testing dataset.

C. Deep Learning Models

In addition to the newly proposed CNN-Attention network, another two deep learning models, the Long Short-Term Memory Network (LSTM) and the Sparse Pseudo-input Gaussian processes (SPGP) were also included in this study for comparison. The CNN-Attention network and LSTM were built on the Pytorch framework and trained on the same GPUs (a Titan V), while the SPGP was implemented in MATLAB 2010b (The MathWorks Inc., USA).

1) CNN-Attention Network:

a) *Multi-scale Convolution Structure*: Traditional convolution operation with a fixed filter size could not extract sufficient EMG information because of the small receptive field. Therefore, in this study, a convolution operation with filters of different sizes was adopted to extract both large-scale and small-scale features simultaneously. Specifically, two multi-scale convolution modules were used in this work, as shown in Fig. 2. Each multi-scale convolution module consists of 3 parallel paths, the filter sizes corresponding to different paths were 3, 5, and 7, respectively. Herein we set the number of multi-scale convolution modules and the parallel paths by tradeoff between the prediction accuracy and the computational complexity [24]. It should be noted that appropriate padding operation is applied into convolution to have its input and output with the same sequence length. To compress the feature sequence length and increase the receptive field of convolution, the output of first multi-scale convolution module was connected with an average pooling layer.

b) *Multi-head Attention Structure*: Afterwards, a multi-head attention structure was utilized to estimate the finger joint angles, for the attention mechanism showed superiority not only in prediction accuracy for long time-series, but also significantly less time to train in language translation tasks [20]. In current study, we stacked three multi-headed self-attention modules ($n = 3$), each multi-headed self-attention module consists of three single-headed attention ($m = 3$). As attention only cannot distinguish the order relationship of the input signal, it is necessary to add positional encoding prior to the input signal.

For an input sequence of the attention module is $X = \{x_0, x_1, \dots, x_n\} \in R^{n \times 1}$, we firstly used information $P = \{p_0, p_1, \dots, p_n\} \in R^{n \times 1}$ to encode the relative or absolute position in sequence X , yielding a new input $C = X + P$. In single-head attention of multi-head attention, the input c_t was linearly mapped to three different spaces, $W_Q \in R^{l \times d_q}$, $W_K \in R^{l \times d_k}$, and $W_V \in R^{l \times d_v}$. Then the attention score function $S(c_t, c_o)$ to represent the weight coefficient of time t and time o was calculated as:

$$S(c_t, c_o) = \frac{(c_t W_Q)(c_o W_K)^T}{\sqrt{d_z}} \quad (1)$$

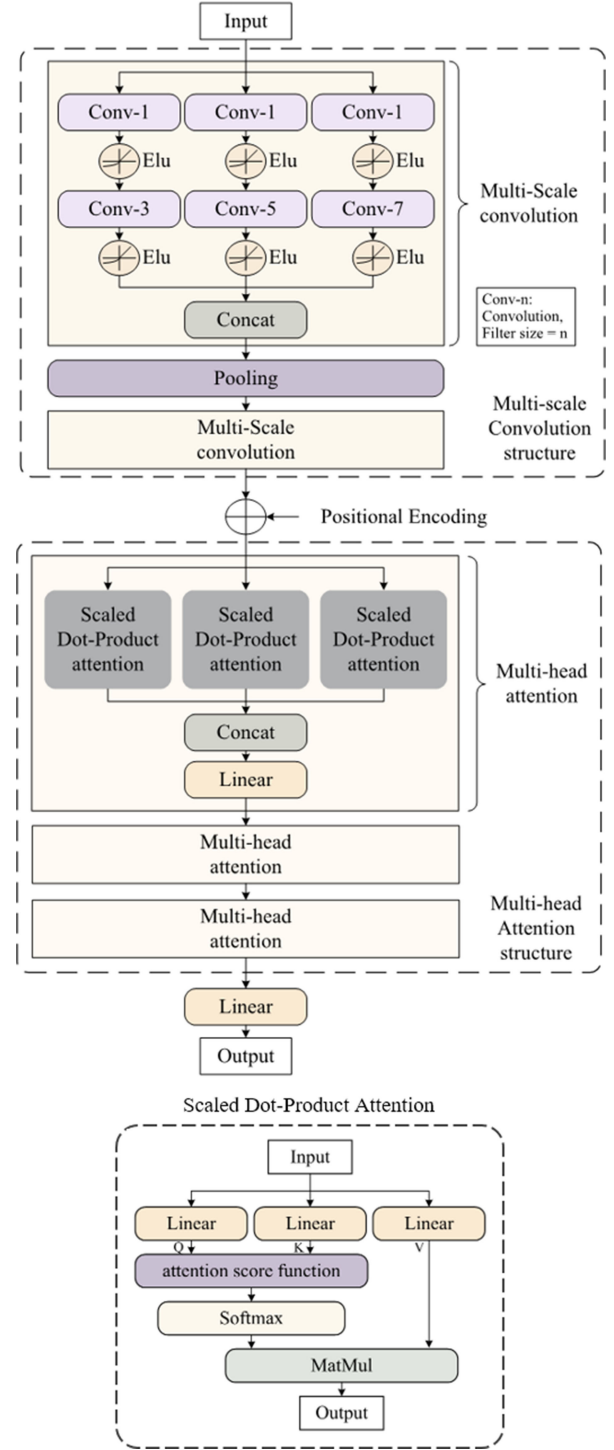


Fig. 2. Architecture of CNN-Attention Network.

Attention score of o -th input vector and i -th input vector is $((x_i + p_i)W_Q)((x_o + p_o)W_K)^T$. After expanding the formula, $p_i W_Q W_K^T p_o^T$ can be seen as a function of the distance between o and i . W_Q and W_K^T are parameters that can be learned in training, $p_i p_o^T$ is a fixed value only related to the relative position between o and i :

$$p_i p_o^T = \sum_{d=0}^{\frac{l}{2}-1} \left[\sin\left(t/1000^{\frac{2d}{l}}\right) \sin\left(o/1000^{\frac{2d}{l}}\right) \right]$$

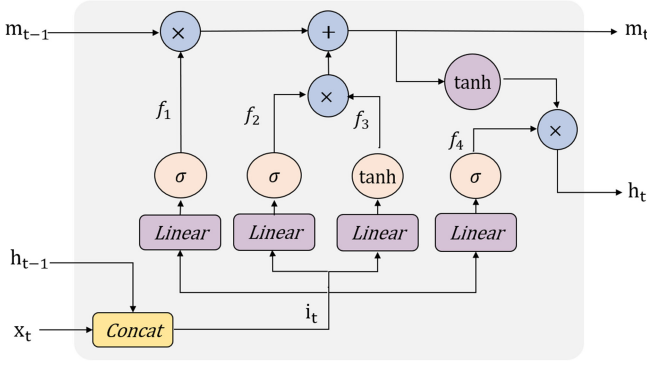


Fig. 3. Structure of LSTM Network.

$$+ \cos\left(t/1000^{\frac{2d}{t}}\right) \cos\left(o/1000^{\frac{2d}{t}}\right)] \quad (2)$$

So, formula (2) can be transformed into formula (3), it is a position dependent formula.

$$p_i p_o^T = \sum_{d=0}^{\frac{l}{2}-1} \cos\left((o-t)/1000^{\frac{2d}{t}}\right) \quad (3)$$

After training, $p_i W_Q W_K^T p_o^T$ enables the attention mechanism to perceive the sequence position information. The output of attention mechanism defined as $H = \{h_0, h_1, \dots, h_n\} \in R^{n \times d_v}$, then output point h_t corresponding to input point x_t can be expressed as:

$$h_t = att(x_t) = \sum_{o=1}^n \frac{s(x_t, x_o) x_o W_V}{\sum_{j=1}^n s(x_t, x_j)} \quad (4)$$

In actual use, the attention mechanism will calculate a query matrix $Q = \{q_0, q_1, \dots, q_n\} \in R^{n \times d_q}$, where $q_i = c_i W_Q$, key vector matrix $K = \{k_0, k_1, \dots, k_n\} \in R^{n \times d_k}$, $k_i = c_i W_K$ and value vector matrix $V = \{v_0, v_1, \dots, v_n\} \in R^{n \times d_v}$, $v_i = c_i W_V$ in parallel for a complete sequence signal. The output vector sequence can be described as:

$$H = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

In above formulas, the d_q , d_k and d_v were set as 64, 64 and 128, respectively. Finally, the Concat module concatenated the outputs H1, H2, H3 of each head attention together, so the output of Multi-Head Self-Attention can be described as:

$$Z = \text{Concat}(H_1, H_2, H_3) W_O \quad (6)$$

2) *Long Short-Term Memory Network (LSTM)*: For comparison, the LSTM which was firstly proposed by Hochreiter and Schmidhuber was also included in this study [25]. It is a type of many-to-many Recurrent Neural Network (RNN) built to model temporal sequences and avoid long-term dependence problems. Compared with RNN, LSTM has been proved to be more suitable for regression-based problems, especially with input data containing long term time information [26]. An overview of LSTM architecture which has 4 interactive layers in each cell was used in this study (Fig. 3). We used five stacked LSTM to increase the depth of the model for improving the fitting ability of the model. The hidden dimension of each layer was set to

256, and the dropout rate was set as 0.1 to prevent over fitting, the last layer outputs 10 joint angles through linear mapping.

3) *Sparse Pseudo-input Gaussian Processes (SPGP)*: Gaussian process regression (GPR) models are nonparametric kernel-based probabilistic models, it has been successfully used for non-linear non-parametric regression and classification, especially in high-dimensional space [27]. Given that the computational complexity and time of the standard GPR would be too large for most practical applications, the Sparse Pseudo-input Gaussian processes (SPGP) was proposed by Edward Selsdon *et al.* in 2006 [28]. For brief please refer to [28] for the code we utilized in this work.

D. Performance Metrics for Evaluation

For each of the three aforementioned deep learning models, the predicted joint angle curves and their corresponding actual joint angle curves were firstly plotted in the same Fig. to visualize the evaluation performance. Moreover, three performance metrics, i.e., the Pearson correlation coefficient (CC), the root mean square error (RMSE), and the coefficient of determination (R2) were used, respectively [14], [29], [30]. CC is commonly used to measure whether two sequences are on a line or not, and to measure the linear relationship between distance. Here, we used it to evaluate how closely the predicted joint angle fits the corresponding actual joint angle. Its calculation formula is as follows:

$$CC = \frac{\sum_{i=1}^N (\theta_{est} - \bar{\theta}_{est}) (\theta_{real} - \bar{\theta}_{real})}{\sqrt{\sum_{i=1}^N (\theta_{est} - \bar{\theta}_{est})^2} \sqrt{\sum_{i=1}^N (\theta_{real} - \bar{\theta}_{real})^2}} \quad (7)$$

where θ_{est} , $\bar{\theta}_{est}$, θ_{real} , and $\bar{\theta}_{real}$ presents the value of estimated joint angle, the mean value of estimated joint angles, the value of real joint angle and the mean value of real joint angles, respectively. The closer the CC value is to 1, the more similar the predicted finger trajectory is to that of the actual movement, and the higher accuracy of the estimation can reach.

RMSE can reflect the deviation between two variables. Here, we used RMSE to evaluate the deviation between the estimated value and the measured value of each joint angle, in degrees (°). RMSE is calculated as follows:

$$RMSE = \sqrt{\sum_{i=1}^N \frac{(\theta_{est} - \theta_{real})^2}{N}} \quad (8)$$

R2 is a comprehensive evaluation index ranging from 0 to 1. It is defined as the percent variability in the true values explained by the estimated values. The larger the R2 value is, the better the estimation performance would be. Here, we used R2 to measure the variance between the estimated value and the measured value of each joint angle.

$$R^2 = 1 - \frac{\sum_{i=1}^N (\theta_{real} - \theta_{est})^2}{\sum_{i=1}^N (\theta_{real} - \bar{\theta}_{real})^2} \quad (9)$$

E. Statistical Analysis

To examine the differences between three deep learning algorithms, the Friedman test, a non-parametric alternative to the one-way ANOVA with repeated measures was used in this

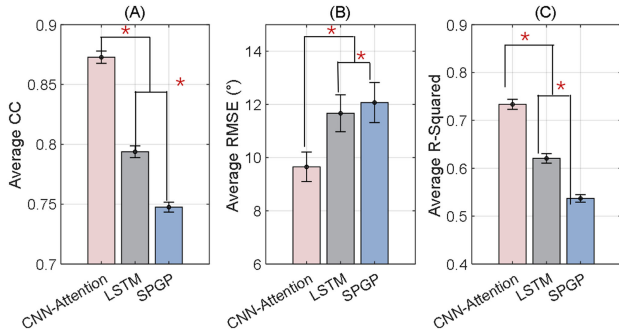


Fig. 4. Overall performance of finger joint angles estimation. The CC, RMSE, and R2 were used to evaluate the (A) CNN-Attention, (B) LSTM, and (C) SPGP, respectively.

study. The CC, RMSE, and R2 were separately used as the dependent variables. Once the Friedman test revealed a statistically significant effect, post-hoc comparisons were performed using the Wilcoxon sign-rank test, with the p-value adjusted by the Bonferroni correction for multiple comparisons. In this work p-value < 0.05 was taken as the threshold of statistical significance.

III. RESULTS

The overall performance of finger joints estimation when using each of the three deep learning models was shown in Fig. 4, where the CC, RMSE, and R2 were averaged across all joint angles and all subjects. It can be seen that the average CC corresponding to CNN-Attention, LSTM, and SPGP were 0.87 ± 0.01 , 0.79 ± 0.01 , and 0.75 ± 0.01 , respectively. The average RMSE corresponding to CNN-Attention, LSTM, and SPGP were 9.65 ± 0.55 , 11.67 ± 0.69 , and 12.07 ± 0.75 , respectively. And the average R2 corresponding to CNN-Attention, LSTM, and SPGP were 0.73 ± 0.01 , 0.62 ± 0.01 , and 0.54 ± 0.01 , respectively. The statistical analysis proved that the proposed CNN-Attention model outperforms the other two deep learning models in terms of CC, RMSE, and R2 (p-value < 0.05), and the LSTM outperforms SPGP (p-value < 0.05).

Fig. 5 shows the estimation accuracy with respect to different joint angles and different subjects, respectively. It can be seen that in Fig. 5(A)–(C), the CNN-Attention provides high CC (>0.82), low RMSE (<12.81) and high R2 (>0.65) for each joint angle, while LSTM and SPGP perform distinctly worse in comparison. By comparing the accuracy with respect to different subjects (Fig. 5(D)–(F)), it can be found that the CNN-Attention maintained high accuracy among all subjects with CC > 0.83. As for RMSE and R2, CNN-Attention also presents superior performance for each subject. These outcomes indicate that the proposed CNN-Attention has the capability, stability and versatility to be applied on various subjects and joint angles. The statistical analysis with respect to each joint and subject consistently showed that the proposed CNN-Attention model outperforms the other two deep learning models regardless of the performance metrics (p-value < 0.05).

Fig. 6 illustrates an example of the estimated joint angles (in red) against measured curves (in blue) when using each of these three models. It is obvious that the CNN-Attention performed

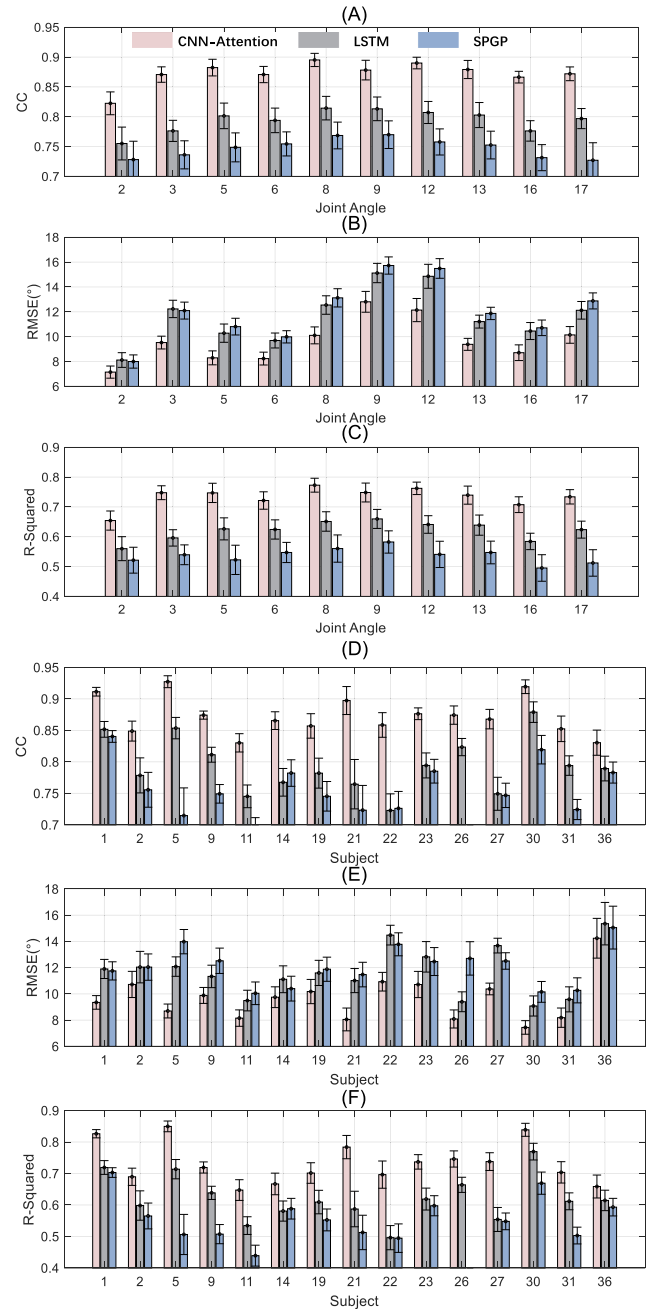


Fig. 5. Estimation accuracy with respect to different joints (A–C) and different subjects (D–F).

markedly better than LSTM and SPGP, because its predicted curves were mostly similar to the measured curves without distortion. The LSTM model also provided accurate trend, but it became unstable when a large change in the joint-angle occurred for four joint angles, 6, 9, 13, and 17. In contrast, the SPGP performed particularly badly, it was barely able to match the pattern of any of the ten measured joint angles, and the distortion became larger when estimating the sharp tuning of for joint angles 6, 9, 13, and 17.

In addition to the accuracy of joint angle estimation, we also compared the computational cost to build a CNN-Attention model and a LSTM model, which were implemented in the same GPUs. As shown in Table I, the CNN-Attention converged faster

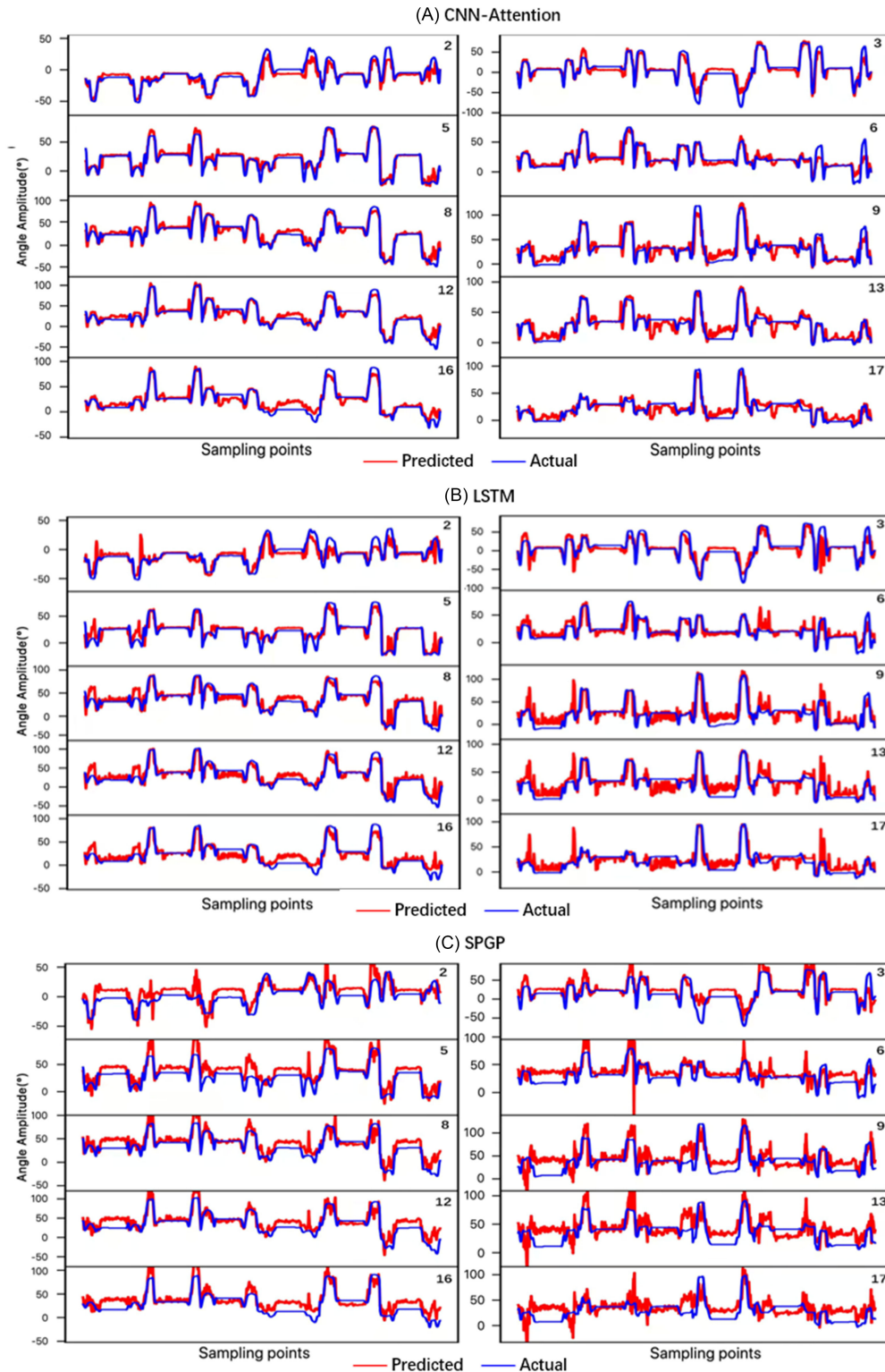


Fig. 6. Estimated joint angles against measured curves for Subject 05, when using (A) CNN-Attention, (B) LSTM, and (C) SPGP.

than LSTM for all subject except for Subject 11, 21, and 27. The average number of epochs to converge was around 100 and 162 for CNN-Attention and LSTM, respectively. The average total time to build a CNN-Attention and LSTM model was 43.00 min and 73.40 min, respectively. These results showed obviously lower computational cost to build a CNN-Attention model than to build a LSTM model.

IV. DISCUSSION

Aiming at improving the accuracy of continuous and proportional estimation of finger movements and accelerating the model training, a new deep learning approach named CNN-Attention was proposed in this study. To examine its performance by comparison, two commonly used methods, the LSTM and SPGP were also included. Their performances

TABLE I
COMPARISON OF COMPUTATIONAL TIME TO BUILD CNN-ATTENTION
AND LSTM

Subj No.	CNN-Attention		LSTM	
	Epochs number	Training time (min)	Epochs number	Training time(min)
01	61	28	194	81
02	106	50	166	72
05	110	42	128	54
09	100	41	266	113
11	128	53	108	47
14	75	38	194	85
19	82	40	132	60
21	216	85	214	92
22	80	38	82	47
23	51	25	172	73
26	90	35	106	43
27	118	55	118	66
30	160	64	200	118
31	60	23	162	69
36	61	28	194	81
AVE $\pm$	99.87 $\pm$	43.00 $\pm$	162.40 $\pm$	73.40 $\pm$
STE	11.30	4.25	12.83	5.81

were evaluated on sEMG and finger joint angles of six grasping movements acquired from 15 subjects.

To accurately describe the estimation performance of each algorithm, the CC, RMSE, and R2 were separately measured between the actual and predicted joint angles. Results shown in Figs. 4 and 5 indicated that the newly proposed CNN-Attention approach significantly outperformed the LSTM and SPGP for its both higher accuracy and better stability across all subjects and all joint angles. While the LSTM and SPGP showed heterogeneity of joints and participants, i.e., LSTM performs worse than SPGP on a few joints and participants. The possible reason leading to the superiority of CNN-Attention maybe that the multi-scale convolution obtains rich features of different scales [5], and the attention structure with constant level path distance make it easier to learn long-range dependencies of time series [20]. The unique recurrent structure of LSTM may account for its superiority over SPGP on most joint angles [16].

The advantage of the proposed CNN-Attention approach was further examined by comparing the predicted curves to actual curves (Fig. 6). It can be found that both LSTM and SPGP are able to detect the trend of the actual finger motion clearly, but the amplitude discrepancies became larger, especially during intervals of sudden inflections at distal joint angle 6, 9, 13, and 17. The relationship between these distal joint angles and sEMG is relatively weaker, which might be an important reason for the degraded accuracy. In contrast, CNN-Attention provides more continuous, stable and accurate joint angle curves to match the actual curves even for these distal joints, which proves again the superiority of CNN-Attention model.

In this work, we also found that the superiority of algorithms is related with performance metrics. For instance, the LSTM was proved superior to SPGP in terms of CC and R2 for all joints, but worse than SPGP for joint 2 and 3 when using RMSE (Fig. 5(B)). Similar conflicting results were found for estimation accuracies among all participants, as shown in Fig. 5(D)–(E). The CC and RMSE evaluate how two curves fit each other from two different perspectives, the waveform similarity and distortion, therefore, inconsistent outcomes could be generated

as a consequence. In this work, the inconsistent results about the superiority of LSTM versus SPGP implies the importance of applying multiple performance metrics.

In addition to estimation accuracy, the computational cost to build a CNN-Attention model was also evaluated and compared with that of LSTM in this study. The results indicate that lower computational cost was needed to build a CNN-Attention model with average time to converge around 43.00 min, obviously shorter than that to build a LSTM model (Table I). The possible reasons are stated as follows. Firstly, the attention mechanisms allow us to parallelize the operations and greatly accelerate a model's training time [20]. Secondly, the CNN-attention architecture has 224010 training parameters, less than that of LSTM (602378), the network with smaller training parameters tends to converge faster. Thirdly, the multi-scale convolution strategy could improve the representation of sEMG feature, which is beneficial to accelerate the training speed. As for a few subjects (Subject 11, 21, and 27) whose computational cost did not reduce when using the CNN-Attention, a possible reason is that the poor data quality for these subjects need longer time to reach convergence.

In this work, there are also some limitations. Firstly, the proposed CNN-Attention was only evaluated on six basic grasp movements, which are far from covering complicated hand movements in our daily life. To develop a more practical deep learning model involving more functional movements, improving position coding by means of relative position embedding in the attention structure would represent the time relationship of multiple grasping movements better [31], [32]. Secondly, the Ninapro database with sparse electrodes was adopted in this study, which could not provide sufficient temporal either spatial information to predict more and finer change of kinematic signals, particular for the variation of distal joint angles. In our ongoing work, utilizing high-density sEMG via flexible and stretchable electrodes that placed on both forearm and hand will be considered, for which could record more finer muscular activities because of the better muscle-electrode contact in comparison with the commonly used rigid electrodes [33]. Thirdly, the optimal window configures which would affect the online performance significantly needs to be intensively investigated in the future work. Lastly, only offline performance of three deep learning approaches were investigated in this work, it is unclear if these metrics could be used as predictors of online usability [34]. Because in practical scenarios, several factors such as the shift in electrode position, variation of muscle contraction force, and arm position would affect the sEMG patterns [35]–[37]. Therefore, evaluating the real-time performance of these algorithms by accelerating their computation in further and improving their robustness against aforementioned factors need to be conducted in the future work.

V. CONCLUSION

Towards accurate estimation of continuous finger kinematics, a novel deep learning approach named CNN-Attention was proposed and evaluated on Ninapro database in this study. Compared with two previously used approaches, LSTM and SPGP, the CNN-Attention showed significantly better accuracy regardless of the performance metrics. Also, it has the capability, stability and versatility to be applied on various subjects and

joint angles. In addition, the computational time to build a CNN-Attention proved to be obviously less than to train a LSTM model. These outcomes suggest that the CNN-Attention is expected to play an important role in the human-machine interaction and cooperation in future.

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